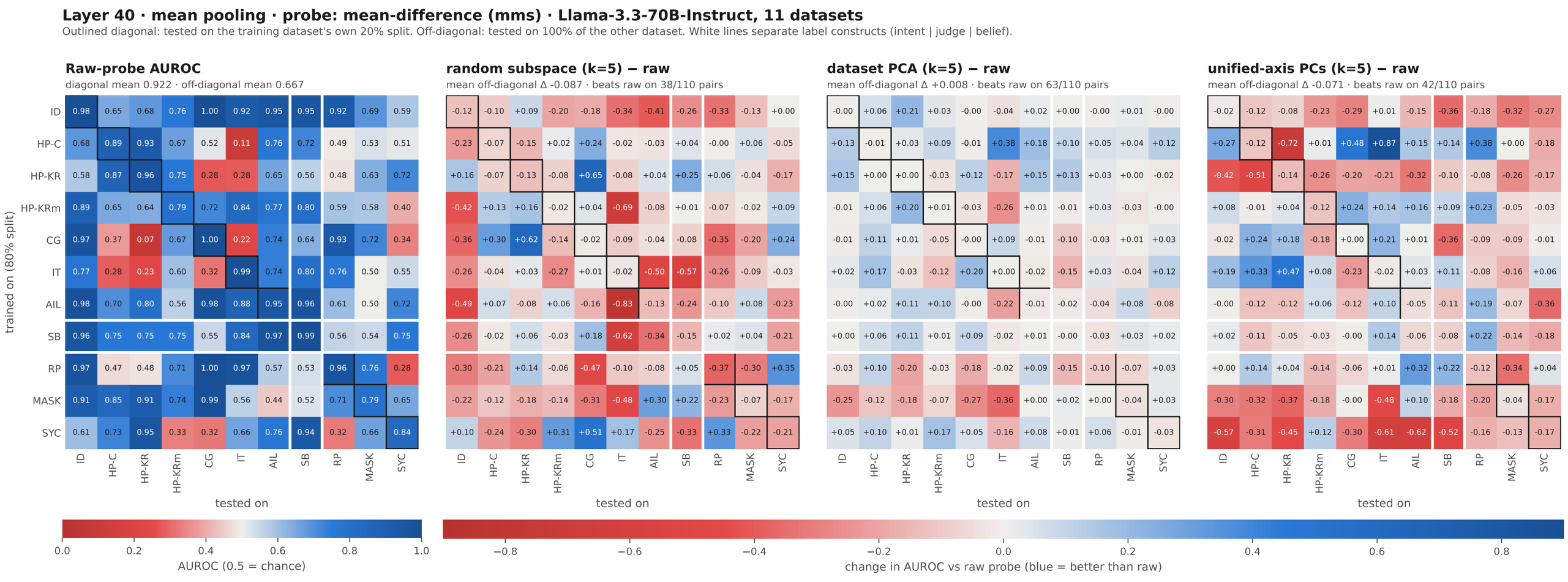

Layer 40 · mean pooling · probe: mean-difference (mms) · Llama-3.3-70B-Instruct, 11 datasets
Outlined diagonal: tested on the training dataset's own 20% split. Off-diagonal: tested on 100% of the other dataset. White lines separate label constructs (intent | judge | belief).
Raw-probe AUROC
diagonal mean 0.922 · off-diagonal mean 0.667
random subspace (k=5) − raw
mean off-diagonal Δ -0.087 · beats raw on 38/110 pairs
dataset PCA (k=5) − raw
mean off-diagonal Δ +0.008 · beats raw on 63/110 pairs
unified-axis PCs (k=5) − raw
mean off-diagonal Δ -0.071 · beats raw on 42/110 pairs
trained on (80% split)
ID HP-C HP-KR HP-KRm CG IT AIL SB RP MASK SYC
tested on
0.0 0.2 0.4 0.6 0.8 1.0
AUROC (0.5 = chance)
−0.8 −0.6 −0.4 −0.2 0.0 0.2 0.4 0.6 0.8
change in AUROC vs raw probe (blue = better than raw)